MILITARY SPECIFICATION

CHARGE, PROPELLING, 8 INCH, M188E1 LOADING, ASSEMBLING, AND PACKING

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 This specification covers the manufacture of Charge, Propelling, M188E1, for 8 Inch Howitzer, M110A1E1 by manufacturing Increment 9 and assembling Increment 9 to the Increment 8 Loading Assembly (dwg 9277175) and Flash Reducer Assembly (dwg 9299188) available only from the M188 propelling charge (MIL-C-48143) production line. To produce M188E1 propelling charges to this specification all component weights required for the Charge, Assembly Weight (3.3.1) must accompany the Increment 8 Loading Assembly from the M188 propelling charge production line. The Increment 8 Loading Assembly is considered acceptable from the M188 propelling charge production line only if the days production of M188 propelling charges from which the Increment 8 Loading Assemblies and Flash Reducers were taken are accepted.

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKAGE DRAWINGS

9312720 - Packing and Marking for Container, Ammo, Metal PA66 for Charge Propelling 8 in. M188E1 for How. SP, M110A1E1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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9312712 - Charge, Propelling, 8 Inch, M188E1
for 8 Inch Howitzer, SP, M110A1E1

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling Charge.-The propelling charge shall comply with all requirements specified on drawing 9312712 and with all requirements specified in applicable specifications.

3.3 Propellant Weight.-The Increment 9 propellant weight shall comply with the weight specified in the loading authorization within plus or minus 0.20 ounce.

3.3.1 Charge, Propelling Assembly Weight.-The charge, propelling assembly weight shall not differ by more than 1.0 ounce from the total of the amount specified in the loading authorization plus tare weight of the empty increments, flash reducer, lacing cord, lacing jacket, and loaded base igniter and tube assembly. (See 6.8).

3.4 Dimensional Gage.

3.4.1 Maximum Length and Diameter.-The completed charge shall freely enter the prescribed gage. The gage shall be designed to inspect the maximum diameter and length. If the diameter feature is separate from the length feature, the diameter feature must extend approximately the full length of the charge.

3.4.2 Minimum Diameter.-The base end of the completed charge shall not freely enter the prescribed gage greater than two inches.

3.5 Workmanship.-All components shall be free of dirt, grease and other foreign material. There shall be no discoloration or deterioration of cloth and no evidence of rips, tears, or other cloth defects. All work performed shall be performed in a thorough workmanlike manner.

3.6 Proving Ground.-The muzzle velocity, pressure, and velocity standard deviation shall be as specified in Table I. (See 6.12)

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TABLE I

Zone	Proj.	Temp	Velocity (feet per second) Minimum	Velocity (feet per second) Maximum	Velocity Standard Deviation FPS	Individual Maximum Pressure PSI
Chg. 9	M106	70°F	2515	2545	6.5	42,100
Chg. 9	M106	145°F (Differential Pressure)				48,100
Chg. 9	M106	-50°F (Differential Pressure)				NA

3.6.1 Differential Pressure.-No individual maximum differential pressure shall exceed 7000 psi. If the mean plus three (3) times the standard deviation exceeds 5000 psi, the lot will be referred (see 6.10). If the mean plus three (3) times the standard deviation exceeds 7000 psi, the lot will be rejected.

3.7 Assembly Torque.-The test plug in the packing container shall be assembled to withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing, without moving.

3.8 Igniter Tubes.-All tubes shall be free of cracks, dents, creases, tears, flat areas, and softness greater than 0.25 inch.

3.9 First Article Inspection.-This specification contains technical provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

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4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a First Article Sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of twenty-four (24) propelling charges packed in metal containers which have been produced by the contractor and selected randomly from the first week's regular production.

4.3.2 Inspection to be Performed.-See MIL-A-48078.

4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of propelling charge assemblies shall contain:

a. Propellant from not more than one lot number from one manufacturer.

b. Increment 8 loading assemblies from not more than one lot number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312729
4.4.2.1	Body				NEXT HIGHER ASSEMBLY 9312726
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cloth improper			3.1	4.5.3
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Liner Assembly				9312728
					NEXT HIGHER ASSEMBLY
					9312726
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Established length		0.40%	3.2	Scale
102	Established width		0.40%	3.2	Scale
103	Area coverage of additive coating, minimum		0.40%	3.2	Visual-Scale
104	Weight of additive		0.40%	3.2	Balance
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312724
4.2.2.3	End, Increment 9				NEXT HIGHER ASSEMBLY 9312725
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cloth improper			3.1	4.5.3
Minor					
201	Established length		0.65%	3.2	Scale
202	Established width of straight section		0.65%	3.2	Scale

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Increment 9, End Assembly				9312725
					NEXT HIGHER ASSEMBLY
					9312726
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Seam improper		0.40%	3.2	Visual-Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Increment 9 Assembly				9312726
					NEXT HIGHER ASSEMBLY
					9312727
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Increment 9 end assembly improperly assembled to body		0.40%	3.2	Visual
102	Any seam incompletely stitched		0.40%	3.2	Visual-Manual
103	Liner assembly improper, improperly assembled missing or improperly located		0.40%	3.2	Visual-Scale
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312727
4.4.2.6	Increment 9 Loading Assembly				NEXT HIGHER ASSEMBLY 9312712
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propellant weight			3.3	4.5.1
102	Any seam incompletely stitched		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312731
4.4.2.7	Jacket, Lacing				NEXT HIGHER ASSEMBLY 9312712
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Distance between rows		0.40%	3.2	Scale
102	Cloth improper			3.1	4.5.3
103	Lacing cord missing or improperly assembled (2 locations)		0.40%	3.2	Visual-Manual
Minor					
201	Jacket of excessive length		0.65%	3.2	Scale
202	Marking missing, misleading, or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.5	Visual
204	Any seam improper or incompletely stitched		0.65%	3.2	Visual-Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Charge, Propelling, 8 Inch, M188E1 for 8 Inch Howitzer, SP, M110A1E1	1 OF 2	9312712
			NEXT HIGHER ASSEMBLY —

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Assembly damaged or improper to the extent that black powder can escape		100%	3.2	Visual-Manual
2	Base pad not in true position with respect to outer diameter of charge (to be determined after straps are securely tied and charge is laced)		100%	3.2	Gage
3	Black powder missing in basepad		100%	3.2	Manual
	Proving Ground				
4	Pressure	(a)		3.6	4.5.4
Major					
101	Dimensional gage failure		100%	3.4	Gage
102	Lacing missing, improper or insecure		100%	3.2	Visual-Manual
103	Igniter protector cap missing or improperly attached		0.40%	3.2	Visual
104	Propelling charge assembly weight		100%	3.3.1	4.5.6

NOTES: (a) Samples tested for Muzzle Velocity and Muzzle Velocity Standard Deviation and Differential Pressure (145°F only) will be monitored for pressure.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.8	CONTINUATION SHEET		9312712
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
105	Assembly damaged to the extent that propellant or potassium sulfate can escape (includes any seams or openings incompletely stitched and broken stitches)		100%	3.2	Visual
106	Any component missing or improperly assembled (including flash reducer)		100%	3.2	Visual
107	Straps loose, broken or insecurely tied		0.40%	3.2	Visual-Manual
108	Condition of igniter tube protruding from Increment 8, prior to assembly of Increment 9		100%	3.8	Visual
	Proving Ground				
109	Muzzle velocity and muzzle velocity standard deviation	15		3.6	4.5.4
110	Differential pressure	20		3.6	4.5.4
	Minor				
201	Evidence of poor workmanship		0.65%	3.5	Visual
202	Location of lacing jacket		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Container, Prior to Insertion of Charge				9312720
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Foreign matter or rust in container		0.40%	3.2	Visual
102	Bottom cushion missing		100%	3.2	Visual
103	Charge not tightly wrapped immediately prior to insertion into container (i.e., all charge full diameter points should be in contact with the wrapper).		0.40%	3.2	Visual
Minor					
201	Damaged container (to include dents protruding greater than 1/4" into container)		0.65%	3.2	Visual-Gage (see 6.4)

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9312720
4.4.2.10	Container, Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Long filler above top of charge (except for one allowable torn filler).		100%	3.2	Visual
102	Charge diametrically loose to the extent that a long filler may be completely inserted		100%	3.2	Manual
103	Any filler (if required) or top cushion missing or improper		100%	3.2	Visual
104	Dimension "X",minimum. (In lieu of measuring Dimension "X, the measurement of Dimension "Y" maximum may be taken)		100%	3.2	Gage
105	Filler inserted in container prior to inserting top cushion		0.40%	3.2	Visual
106	More than five long fillers present		0.40%	3.2	Visual
107	Overlap of one long filler with another at top of charge		0.40%	3.2	Visual
108	Cover cannot be removed by hand (immediately prior to application of torque)		0.40%	3.2	Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.10	CONTINUATION SHEET				9312720
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109 Minor	Charge cannot be removed by hand None defined	(a)		3.2	4.5.7

NOTES: (a) One sample from each eight (8) hours production.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312720
4.4.2.11	Container, Sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Test plug missing		100%	3.2	Visual
102	DOD symbol missing, misleading or unidentifiable (if applicable)		0.40%	3.2	Visual
103	Test plug assembly torque		0.40%	3.7	Test (see 6.11)
104	Air pressure test		100%	3.2	4.5.2
Minor					
201	Test plug not greased (to be checked after air pressure test and prior to reassembly)		0.65%	3.2	Visual
202	Metallic seal missing or incorrectly applied		0.65%	3.2	Visual
203	Marking missing, misleading, or unidentifiable		0.65%	3.2	Visual
204	Cover incompletely engaged		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63099 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312720
4.4.2.12	Containerized Charge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
NOTE (a)	Tear down Inspection	(b)		(c)	4.5.5

NOTES:

(a) Category is critical or major dependent upon the category of the defect inspected.
(b) See paragraph 4.4.3.1.
(c) Requirement paragraph is dependent upon inspection performed.

NOTES:

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4.4.3 Testing

4.4.3.1 Teardown Inspection

4.4.3.1.1 Initial Lot Sampling.-Three completely assembled charges shall be randomly selected from each days production of the first production lot until three consecutive days production is found acceptable. Subsequent sampling will be on completely assembled charge from each days production. If there is a lapse in production in excess of ninety (90) days, sampling will return to initial lot sampling.

4.4.3.1.2 Subsequent lot sampling.-One completely assembled charge shall be randomly selected from each days production.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.3.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 of this specification.

4.5 Test Methods and Procedures

4.5.1 Propellant Weight.-(see 3.3) -The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. Any increment which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.4.4.

4.5.1.1 Method 1.- The loaded increment will be inspected 100 percent. The weight of the propellant will be determined by subtracting an average weight of the empty increment bag or the actual weight as marked on the individual increment bag from the loaded increment weight.

4.5.1.2 Method 2.- The weight of the propellant will be inspected 100 percent prior to loading the increment on a different balance. In addition, the propellant weight of the loaded increment will be inspected using a sampling plan from MIL-STD-105 or MIL-STD-1235 with an AQL of 0.40 percent. The weight of the propellant will be determined by subtracting an average weight of the empty increment bag or the actual weight as marked on the individual increment bag from the loaded increment weight.

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4.5.2 Air Pressure Test.-Each container shall be tested as specified on the appropriate drawing. A pressure drop equivalent to 3 inches of water or greater shall be cause for rejection and removal from the lot.

4.5.3 Cloth.-At the time cloth is introduced to the sewing or cutting operation, an identification will be made for each roll to verify that proper material is used. Any cloth failing to be identified as proper material in accordance with applicable drawings and specifications shall be rejected from the lot.

4.5.4 Proving Ground.-This test shall be conducted at a Government Proving Ground in accordance with the applicable acceptance test procedure using M106, HE projectiles inert loaded within 0.5 pound of an as fired weight of 200.0 pounds using Primer, M82 and fired in cannon M201 with 95 percent (minimum) remaining life. Firing shall be conducted with a spindle to charge base stand-off of 1 inch (plus or minus 1/4 inch). Prior to firing, all charges shall be temperature conditioned at the required temperature for a minimum of 16 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply to all -50°F, 145°F and 70°F conditioning: -50°F from -50°F to -45°F; 145°F from 140°F to 145°F; 70°F from 68°F to 72°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Velocity coils will be used to measure the muzzle velocity of each charge.

4.5.4.1 Muzzle-Velocity and Muzzle Velocity Standard Deviation.-Determination of acceptance will be in accordance with MIL-STD-414, Section B, Table B3, AQL of 2.50%. All muzzle velocity and muzzle velocity standard deviation samples will be ballistic tested at 70°F. Fifteen (15) charges will be fired for muzzle velocity and muzzle velocity standard deviation. If the sample is determined unacceptable the lot will be rejected. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.76 exceeds the applicable lot requirement, the lot shall be rejected. Fifteen (15) calibration charges will be fired alternately with the test samples and the muzzle velocity and pressure of all the test charges will be corrected to standard conditions.

4.5.4.2 Pressure.-If any propelling charge individual corrected maximum pressure exceeds the applicable requirement, the lot shall be rejected.

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4.5.4.3 Differential Pressure.-Two pressure-time gages at different locations (one in the spindle and one at the projectile base) will be used to record pressure-time traces. Differential pressure traces will be obtained by subtracting the spindle gage output from the projectile gage output. The maximum differential pressure of any charge will be defined as the maximum positive amplitude of the first positive wave of the differential pressure trace prior to peak pressure. Any trace which contains no amplitude in the positive direction will be considered to have a maximum differential pressure of zero. Any sample failing to meet the maximum differential requirements shall be cause for the lot to be referred or rejected (see 3.7.1). Ignition delay will be recorded for information on all differential pressure firings. Ten (10) charges will be conditioned to 145°F and ten (10) charges will be conditioned to -50°F for differential pressure firings. Ten (10) calibration charges will be fired intermixed (five (5) at each temperature) with the test samples and the muzzle velocity and pressure of all test charges corrected to standard conditions.

4.5.5 Teardown Inspection.-Each container and propelling charge will be subjected to each of the critical (except critical defect 2. and 4., paragraph 4.4.2.8) and 100 percent major (except major defect 101, paragraph 4.4.2.6; major defect 104 paragraph 4.4.2.8; major defect 104, paragraph 4.4.2.10) inspections defined in 4.4.2 of this specification. Samples may be accumulated for one week's production, if desired, prior to the actual inspection. The inspection will be performed by the contractor and witnessed by the Government. The detection of any one of the above defects shall be cause for rejection of the charges produced on the day represented by the sample.

4.5.6 Charge, Propelling, Assembly Weight.-The tare weights as marked on the individual assembled base igniter and tube assembly, and empty increment assemblies plus average or actual weight of the flash reducer plus the average tare weight of the lacing cords and jacket together with the nominal propellant weights as specified in the loading authorization shall be subtracted from the assembled charge, propelling, assembly weight. If the resulting weight deviation is greater than the requirement, the propelling charge will be classified defective and removed from the lot. (see 6.9)

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4.5.7 Charge Removal from Container.-Each sample shall be tested for removal of the propelling charge from the container by hand. The charge may be pulled out by the tying straps or the container may be inverted and the charge either pulled out or slid out. Each charge tested shall be checked for damage to the tying straps (looseness, tears, etc.) and the charge will be repaired (retie the tying straps and relace the charge, if necessary). Each charge tested shall be re-wrapped (may use the same wrapper if the wrapper is not badly damaged) and re-packed in accordance with this specification and the applicable drawings. Any charge that cannot be removed from the container using the above method shall be cause for rejection of the production represented by the sample.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.-The propelling charges shall be packed, sealed, tested and marked in accordance with the specified requirements of drawing 9312720.

6. NOTES

6.1 Intended Use.-The propelling charges covered by this specification are intended for use in the 8 Inch M110A1E1 Howitzer, SP.

6.2 Ordering Data.-See MIL-A-48078.

6.3 Submission of Designs for Approval.-See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, Attn: SARPA-QA-T, Dover, New Jersey 07801.

6.4 Visual Examination Qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.5 Inspection Lot Size.-It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

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6.6 Proving Ground.-A copy of all proving ground data should be submitted to Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-A and SARPA-ADEP-3, Dover, New Jersey 07801.

6.7 Teardown Inspection.-A list of all defects found during teardown inspection shall be submitted to Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-A, Dover, New Jersey 07801.

6.8 Charge, Propelling Assembly Weight and Base Igniter and Tube Assembly Weight.-This requirement applies to the load plant at the time of loading only. Due to subsequent change in tare weight of the components because of variation in material and moisture content, this test cannot be repeated after initial manufacture to the degree of accuracy required.

6.9 Charge, Propelling, Assembly Weight.-

Weights To Be Determined		Assembly Drawing Number
Actual Weight of Base Igniter and Tube Assembly	-	9285356
Average/Actual Weight of Flash Assembly	-	9299188
Actual Weight of Body Assembly	-	9277178
Actual Weight of Increment 9 Assembly	-	9312726
Average Weight of Lacing Cord	-	9277173
Average Weight of Lacing Jacket	-	9282014
Nominal Propellant Weight	-	Loading Authorization
Component Weight (sum of above weights)		

The propelling charge assembly weight minus the component weight equals the "Weight Deviation".

6.10 Differential Pressure.-If the differential pressure requires referral of the lot, the disposition of the lot shall be referred to Picatinny Arsenal, Attn: SARPA-QA-A-A, Dover, New Jersey 07801.

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6.11 Test Plug Assembly Torque.-This requirement applies to the load plant at the time of loading only. Due to creep of the mechanical properties of the test plug washer, this test cannot be repeated after initial manufacture.

6.12 Proving Ground Summary (See 4.5.4).-

Test	Zone	Sample Size	Number of Ref Rounds	Velocity	Maximum Individual Pressure	Differential Pressure
70°F	9	15	15	*	42,100	NA
145°F	9	10	5	INFO	48,100	INFO
-50°F	9	10	5	INFO	INFO	INFO

*See 4.5.4.1 and 3.7

Custodian:
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Preparing Activity:
Army-PA

Project Number 1320-A-341

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